

Part 1: Classify each polynomial

Classify each as a monomial, binomial, trinomial, polynomial, or constant.

1. $-x^3 - 4x$

3. $-x^2 - 8$

2. $3x$

4. $-4x^3 - 9x$

Part 2: Standard form

Rewrite each polynomial in standard form.

1. $7x^2 - 8 - 6x^4 - 9x$

2. $-9 + 8x^3 + x^2$

3. $-8x^4 + 4 + 7x - 8x^2$

Part 3: Add polynomials

Add the following polynomials by combining like terms.

1. $(x^2 - 5x + 2) + (-6x^3 - 3x^2 - 6x - 1)$

3. $(-5x^6 + x^4 - 5) + (-8x^4 - 9x + 4)$

2. $(-3x^2 - 2x + 10) + (-4x^3 + 4x^2 + 4x + 3)$

Part 4: Subtract polynomials

Subtract the following polynomials by combining like terms.

1. $(11x^3 - 10x^2 + 3x + 6) - (9x^2 - x - 9)$

3. $(4x^3 + 2x - 1) - (-2x^2 + 8)$

2. $(5x^2 - 9x + 6) - (8x^3 + 12x^2 + 3x + 9)$

Part 5: Find the products

Find the following products. Write your answers in standard form.

1. $5x^3(2x^2 - 4x - 3)$

4. $(x - 1)(x + 4)$

2. $5x(-4x^2 - 2x + 1)$

5. $(2x - 2)(-5x^2 - 4x - 9)$

3. $(2x - 5)(x - 9)$

6. $(2x + 1)(-2x^2 + 7x + 7)$

Part 6: Divide

Divide. Remember to divide every term of the polynomial.

1. $\frac{-3x^3 + 18}{3}$

3. $\frac{3x^6 + 3x^5 + 3x^4}{3x^2}$

2. $\frac{6x^8 - 24x^5 - 42x^4}{6x^2}$

Part 7: Factor

Factor the following polynomials completely.

1. $x^2 - 1$

6. $12x^2 + 13x - 14$

2. $x^2 - 5x - 24$

7. $4x^2 + 4x - 63$

3. $x^2 - 3x - 28$

8. $6x^2 - 31x + 35$

4. $x^2 - 4x + 3$

9. $3x^2 - 20x - 32$

5. $x^2 - 13x + 36$

10. $6x^2 + x - 15$

13. $x^2 - 225$

11. $4x^2 - 41x + 72$

14. $9x^2 - 1$

12. $x^2 - 5041$

Part 8: Exponent properties

Simplify the following using exponent properties. Write your answers with positive exponents.

1. $7x^5y^2 \cdot 5x^7y^7$

6. $\frac{12x^5y^{-4}z^4}{4x^3y^{-6}z^0}$

2. $6a^6b^{-6} \cdot 9a^{-4}b^6$

7. $\frac{3x^4y^{-7}z^8}{3xy^{-9}z^0}$

3. $7x^5y^6 \cdot 8x^5y^5$

8. $9a^2b^{-3} \cdot 5a^{-3}b$

4. $\frac{9x^4y^{-7}z^2}{xy^{-1}z^0}$

9. $11x^4(9xyz^5)^2$

5. $\frac{3x^2y^{-4}z^5}{xy^{-2}z^0}$

Part 9: Radical properties

Simplify the following using radical properties. Assume all variables are positive.

1. $\sqrt[4]{810x^8y^5z^4}$

3. $\sqrt{\frac{270x^2y^7}{3}}$

2. $\sqrt{176x^3y^5z^3}$

4. $\sqrt[3]{72x^3y^4}$

6. $\sqrt{48x^9y^6z^7}$

5. $\sqrt[3]{\frac{264x^{12}y^{12}}{3}}$

7. $\sqrt{16x^4y^6}$

8. $\sqrt[3]{189x^7y^{13}}$

Part 10: Lines

State the slope m and the x - and y -intercepts of each line.

1. $y = 5x - 45$

3. $4y = 4x - 16$

2. $y = 7x + 42$

Part 11: Parabolas

State the vertex of each parabola, then state whether it has a maximum or a minimum.

1. $y = -2x^2 + 36x - 161$

3. $y = 2x^2 - 4x - 5$

2. $y = -x^2 + 20x - 108$

Part 12: Solve by factoring

Solve each equation by factoring.

1. $3x^2 + 17x - 6 = 0$

3. $3x^2 - 11x - 42 = 0$

2. $2x^2 + 7x - 30 = 0$

4. $5x^2 - 8x + 3 = 0$

5. $6x^2 - 13x - 15 = 0$

6. $5x^2 + 31x + 6 = 0$

Part 13: Solve using the quadratic formula

Solve each equation using the quadratic formula.

1. $3x^2 + 10x + 4 = 0$

4. $3x^2 + 10x - 11 = 0$

2. $2x^2 - 3x - 7 = 0$

5. $3x^2 + 9x + 1 = 0$

3. $2x^2 - 9x - 9 = 0$

Part 14: Completing the square

State the values that belong in the blanks to complete the square.

1. $5x^2 + 30x + 6 = 0 \Rightarrow 5(x^2 + 6x + \text{----}) = -6 + \text{----}$

2. $2x^2 + 12x - 7 = 0 \Rightarrow 2(x^2 + 6x + \text{----}) = 7 + \text{----}$

Part 14: Vertex form by completing the square

Change the standard form to vertex form by completing the square.

1. $f(x) = 4x^2 - 32x + 73$

2. $f(x) = 4x^2 - 48x + 145$

3. $f(x) = -2x^2 + 4x + 4$

Answer Key

Part 1: Classify each polynomial

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|-------------|-------------|
| 1. Binomial | 3. Binomial |
| 2. Monomial | 4. Binomial |

Part 2: Standard form

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|----------------------------|---------------------|----------------------------|
| 1. $-6x^4 + 7x^2 - 9x - 8$ | 2. $8x^3 + x^2 - 9$ | 3. $-8x^4 - 8x^2 + 7x + 4$ |
|----------------------------|---------------------|----------------------------|

Part 3: Add polynomials

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|-----------------------------|----------------------------|----------------------------|
| 1. $-6x^3 - 2x^2 - 11x + 1$ | 2. $-4x^3 + x^2 + 2x + 13$ | 3. $-5x^6 - 7x^4 - 9x - 1$ |
|-----------------------------|----------------------------|----------------------------|

Part 4: Subtract polynomials

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|------------------------------|-----------------------------|---------------------------|
| 1. $11x^3 - 19x^2 + 4x + 15$ | 2. $-8x^3 - 7x^2 - 12x - 3$ | 3. $4x^3 + 2x^2 + 2x - 9$ |
|------------------------------|-----------------------------|---------------------------|

Part 5: Find the products

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|----------------------------|----------------------|-------------------------------|
| 1. $10x^5 - 20x^4 - 15x^3$ | 3. $2x^2 - 23x + 45$ | 5. $-10x^3 + 2x^2 - 10x + 18$ |
| 2. $-20x^3 - 10x^2 + 5x$ | 4. $x^2 + 3x - 4$ | 6. $-4x^3 + 12x^2 + 21x + 7$ |

Part 6: Divide

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|---------------|------------------------|----------------------|
| 1. $-x^3 + 6$ | 2. $x^6 - 4x^3 - 7x^2$ | 3. $x^4 + x^3 + x^2$ |
|---------------|------------------------|----------------------|

Part 7: Factor

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|---------------------|------------------------|------------------------|
| 1. $(x - 1)(x + 1)$ | 6. $(4x + 7)(3x - 2)$ | 11. $(4x - 9)(x - 8)$ |
| 2. $(x - 8)(x + 3)$ | 7. $(2x + 9)(2x - 7)$ | 12. $(x - 71)(x + 71)$ |
| 3. $(x - 7)(x + 4)$ | 8. $(2x - 7)(3x - 5)$ | 13. $(x - 15)(x + 15)$ |
| 4. $(x - 3)(x - 1)$ | 9. $(3x + 4)(x - 8)$ | 14. $(3x - 1)(3x + 1)$ |
| 5. $(x - 9)(x - 4)$ | 10. $(2x - 3)(3x + 5)$ | |

Part 8: Exponent properties

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|---------------------|--------------------------|----------------------|
| 1. $35x^{12}y^9$ | 4. $\frac{9x^3z^2}{y^6}$ | 7. $x^3y^2z^8$ |
| 2. $54a^2$ | 5. $\frac{3xz^5}{y^2}$ | 8. $\frac{45}{ab^2}$ |
| 3. $56x^{10}y^{11}$ | 6. $3x^2y^2z^4$ | 9. $891x^6y^2z^{10}$ |

Part 9: Radical properties

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|--------------------------|---------------------------|---------------------------|
| 1. $3x^2yz\sqrt[4]{10y}$ | 4. $2xy\sqrt[3]{9y}$ | 7. $4x^2y^3$ |
| 2. $4xy^2z\sqrt{11xyz}$ | 5. $2x^4y^4\sqrt[3]{11}$ | 8. $3x^2y^4\sqrt[3]{7xy}$ |
| 3. $3xy^3\sqrt{10y}$ | 6. $4x^4y^3z^3\sqrt{3xz}$ | |

Part 10: Lines

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| 1. $m = 5$, $b = -45$,
x -intercept $= (9, 0)$,
y -intercept $= (0, -45)$ | 2. $m = 7$, $b = 42$,
x -intercept $= (-6, 0)$,
y -intercept $= (0, 42)$ | 3. $m = 1$, $b = -4$,
x -intercept $= (4, 0)$,
y -intercept $= (0, -4)$ |
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Part 11: Parabolas

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| 1. $(9, 1)$, maximum | 2. $(10, -8)$, maximum | 3. $(1, -7)$, minimum |
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Part 12: Solve by factoring

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|----------------------------------|----------------------------------|-----------------------------------|
| 1. $x = -6$ or $x = \frac{1}{3}$ | 3. $x = -\frac{7}{3}$ or $x = 6$ | 5. $x = -\frac{5}{6}$ or $x = 3$ |
| 2. $x = -6$ or $x = \frac{5}{2}$ | 4. $x = \frac{3}{5}$ or $x = 1$ | 6. $x = -6$ or $x = -\frac{1}{5}$ |

Part 13: Solve using the quadratic formula

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|---|---|---|
| 1. $x = -\frac{5}{3} - \frac{\sqrt{13}}{3}$ or $x = -\frac{5}{3} + \frac{\sqrt{13}}{3}$ | 3. $x = \frac{9}{4} - \frac{3\sqrt{17}}{4}$ or $x = \frac{9}{4} + \frac{3\sqrt{17}}{4}$ | 5. $x = -\frac{3}{2} - \frac{\sqrt{69}}{6}$ or $x = -\frac{3}{2} + \frac{\sqrt{69}}{6}$ |
| 2. $x = \frac{3}{4} - \frac{\sqrt{65}}{4}$ or $x = \frac{3}{4} + \frac{\sqrt{65}}{4}$ | 4. $x = -\frac{\sqrt{58}}{3} - \frac{5}{3}$ or $x = -\frac{5}{3} + \frac{\sqrt{58}}{3}$ | |

Part 14: Completing the square

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| 1. 9 and 45 | 2. 9 and 18 |
|-------------|-------------|

Part 14: Vertex form by completing the square

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|----------------------------|----------------------------|-----------------------------|
| 1. $f(x) = 4(x - 4)^2 + 9$ | 2. $f(x) = 4(x - 6)^2 + 1$ | 3. $f(x) = -2(x - 1)^2 + 6$ |
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